

1 Introduction

Helios is a solar inverter monitoring system.

It was first deployed in 2021 at the Ningbo plant.

2 Architecture

The system has three tiers: collectors, broker and dashboard.

2.1 Collectors

Each collector polls inverters every 15 seconds over Modbus TCP.

2.2 Message broker

Readings are published to a NATS cluster with 3 nodes.

3 Operations

Operators review alarms twice per shift.

3.1 Alarm thresholds

An over-temperature alarm fires at 78 degrees Celsius.

3.2 Maintenance

Firmware is updated quarterly during the Sunday window.